

# BIOBRAUSE

## About Us

### Moritz Lück

I co-founded BioBrause in 2019, transitioning from LEGO EV3 to advanced custom Raspberry Pi systems. This journey helped me secure a 1st Place victory at the 2024 RoboCup International Rescue Maze. In our current team, I am responsible for the overall software architecture, custom PCB design, AI training, and strategic in-game decision making.

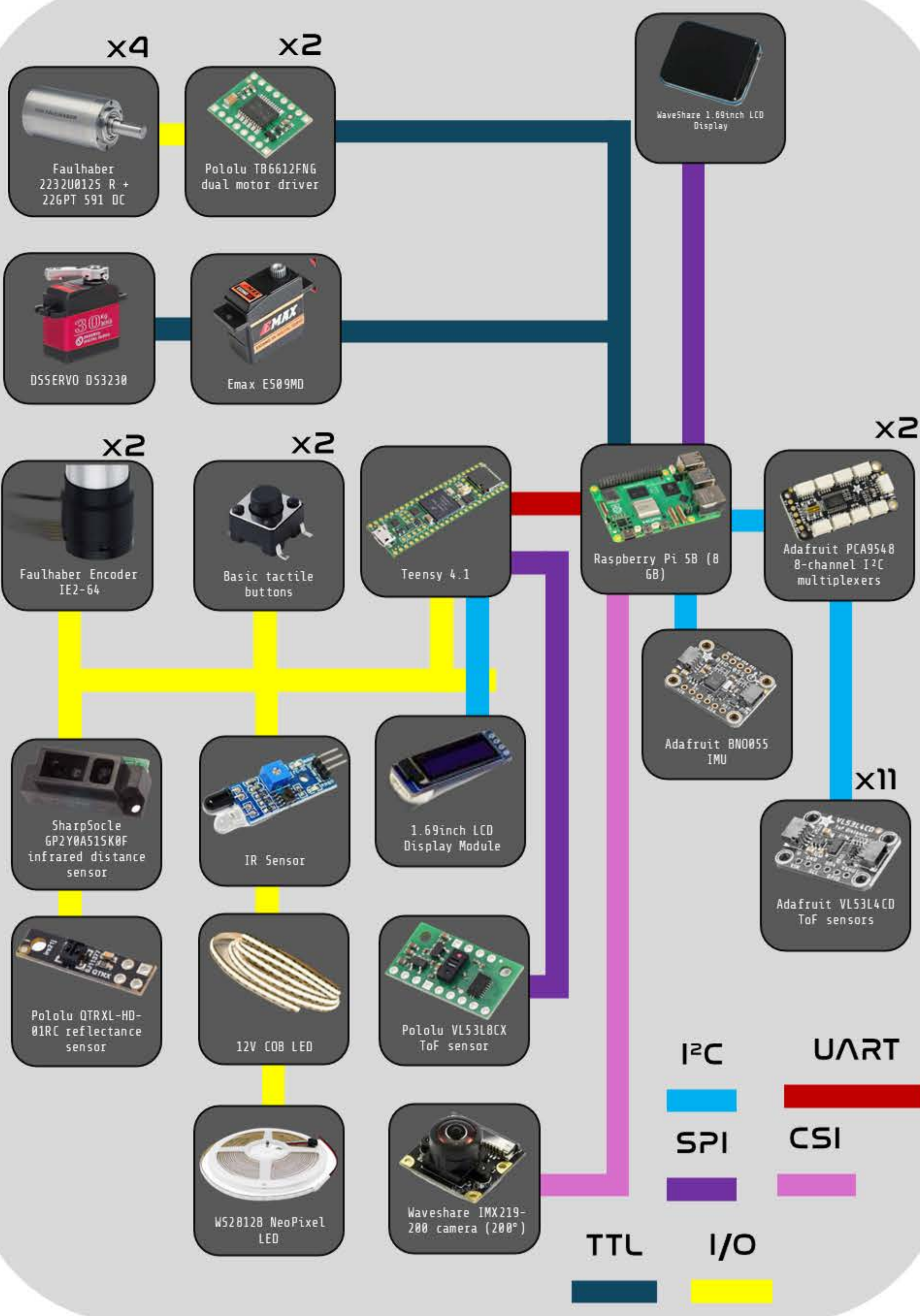
### Abdulsamed Peteja

I joined BioBrause this season after competing in the 2024 and 2025 European RoboCup Championships (Rescue Maze). While my background spans software and hardware, my main focus for this robot was mechanical engineering. I handled the complete Fusion360 CAD design, 3D printing, and manufacturing to maximize our robot's modularity, stability, and grip.

## Our Achievements

- 2025 | Rescue Line
  - European Champion
  - German Vice-Champion
- 2024 | Rescue Maze
  - World Champion
  - German Champion
- 2023 | Maze Entry
  - European Champion
  - German Champion

## Electrical Components



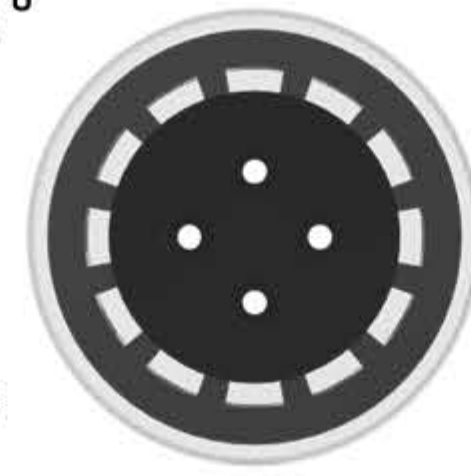
- 2x WaveShare IMX219 (200° FOV):
- Downward for line tracking; forward for victim detection.
- 11x Adafruit VL53L4CD (ToF):
- 360° obstacle avoidance and wall-following.
- Pololu VL53L8CX (8x8 ToF):
- Grid mapping for precise victim targeting.
- Sharp GP2Y0A51SK0F (IR Distance):
- Cross-verifies ToF data to prevent false obstacle detections.
- Adafruit BNO055 (IMU):
- Orientation tracking, precise 90° turns, and ramp detection.
- Pololu QTRXL-HD-01RC:
- Detects the silver Evacuation Zone entrance.
- Internal IR Light Barrier:
- Physically confirms a victim is secured inside the cup.
- Tactile Buttons:
- Robust physical wall alignment and bump detection.

- 4x Faulhaber 2232U0125S:
- Powerful 4x4 direct-drive motors for ramp traversal.
- 4x Faulhaber IE2-64 Encoders:
- Real-time distance and motion feedback.
- DSServo DS3230:
- High-torque servo to rotate the main rescue arm.
- Emax E509MD:
- Tensions the rubber band to trap victims inside the cup.



## Custom Silicone Wheels

To ensure flawless traction on 25° ramps and speed bumps, we manufactured custom 55mm wheels. Consisting of a 3D-printed PLA core and a custom-cast 10A Shore silicone tread, this soft profile eliminates downhill slipping and maximizes grip across unpredictable terrain.

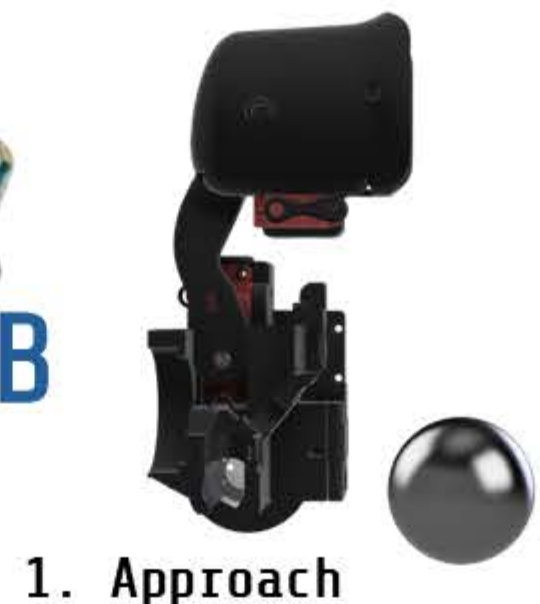


## Custom PCB



League:  
Rescue Line

Country:  
Germany



## Language used



Python

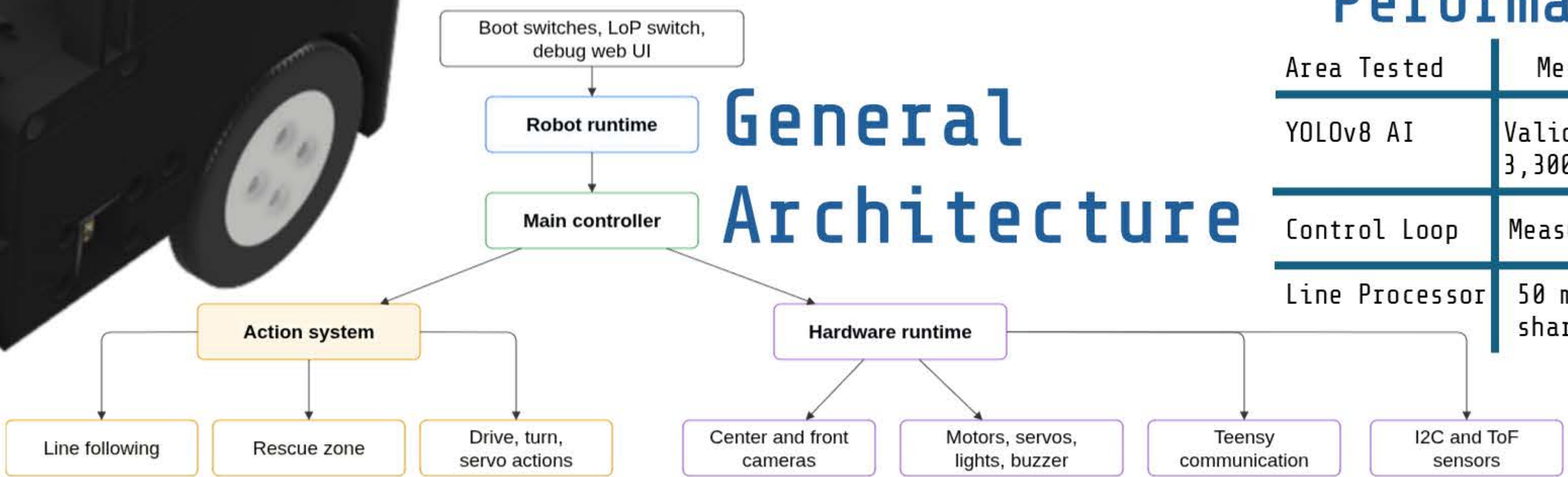


C++

## Rescue Cup

We intentionally avoided complex internal storage to reduce weight and mechanical failure points. Instead, a wide cup lowers over the victim, easily tolerating slight positioning errors. A secondary servo tensions a rubber band to trap the ball. Internally, a light-barrier physically confirms the capture, while conductive metal strips instantly classify the victim as alive or dead before transport.

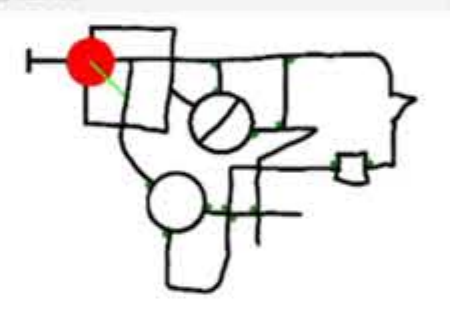
## General Architecture



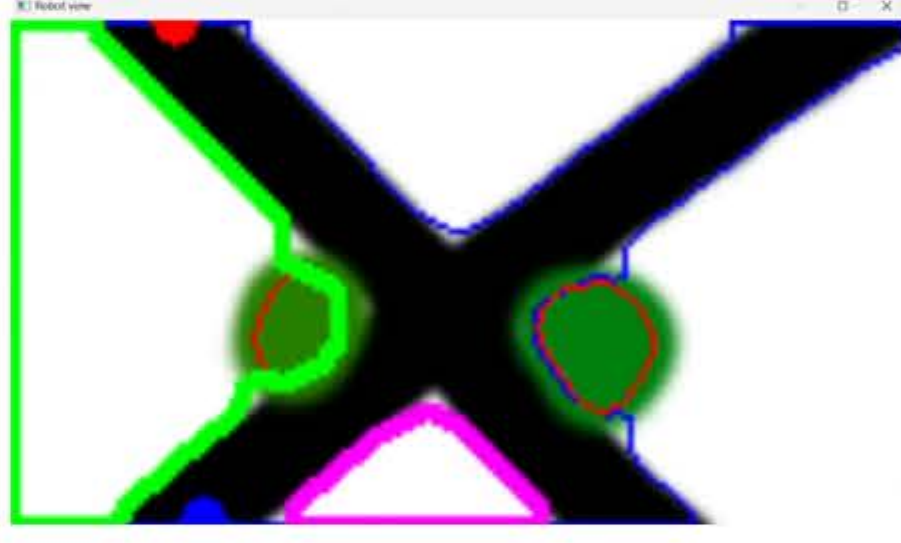
## Performance & Testing Results

| Area Tested    | Method / Metric                   | Result                              |
|----------------|-----------------------------------|-------------------------------------|
| YOLOv8 AI      | Validation on 3,300 custom images | mAP@0.5: 0.995 (High accuracy)      |
| Control Loop   | Measured execution time           | Avg: 1.24 ms (Stable real-time I/O) |
| Line Processor | 50 mixed tiles with sharp bends   | borders ignored; 100% resolved      |

## Custom 2D Simulator



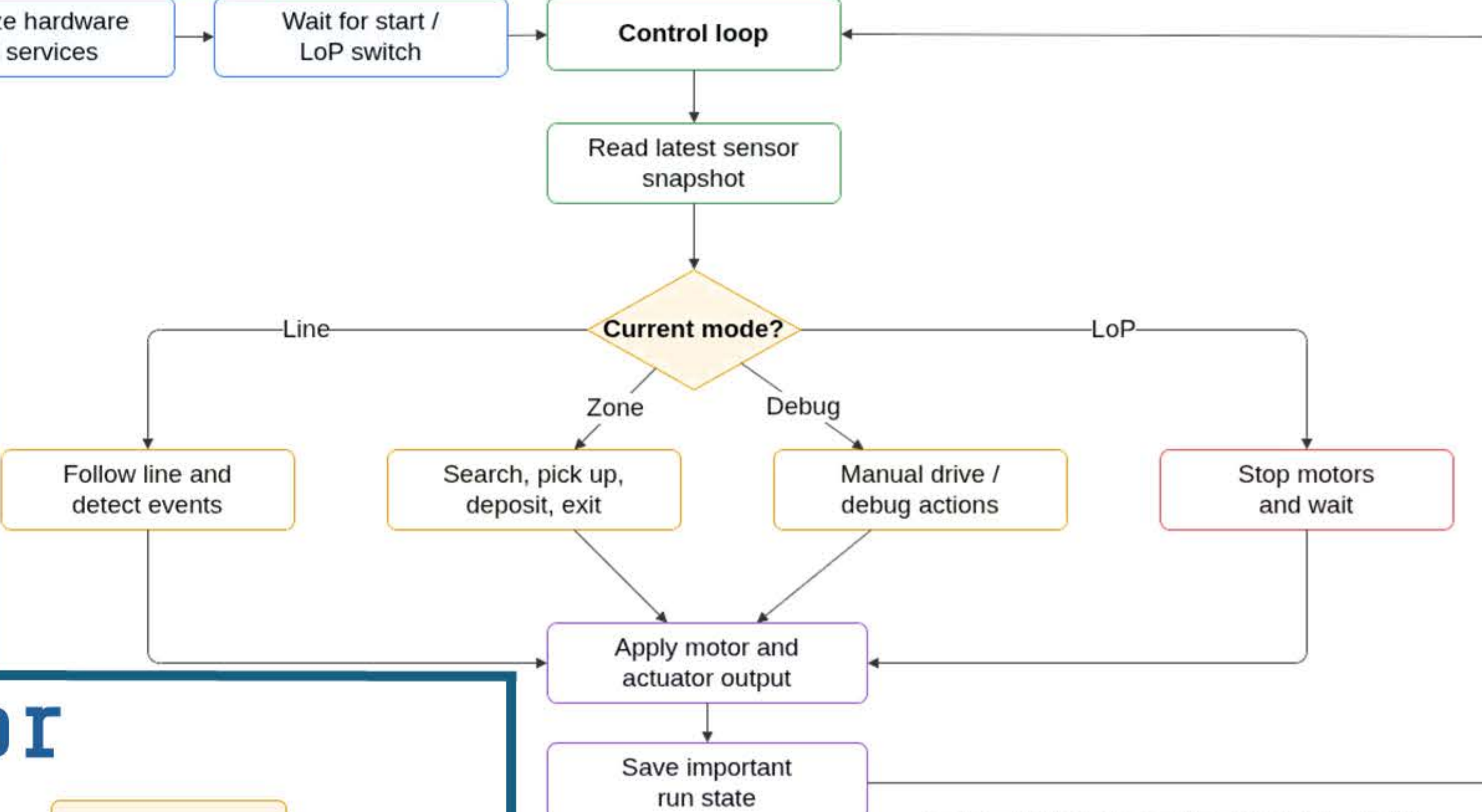
To give us a advantage, we developed a custom 2D simulation environment. It generates a virtual camera image based on a digital track map. The image is processed by the exact same OpenCV pipeline running on the real robot. This allowed us to rapidly reproduce, test, and fix.



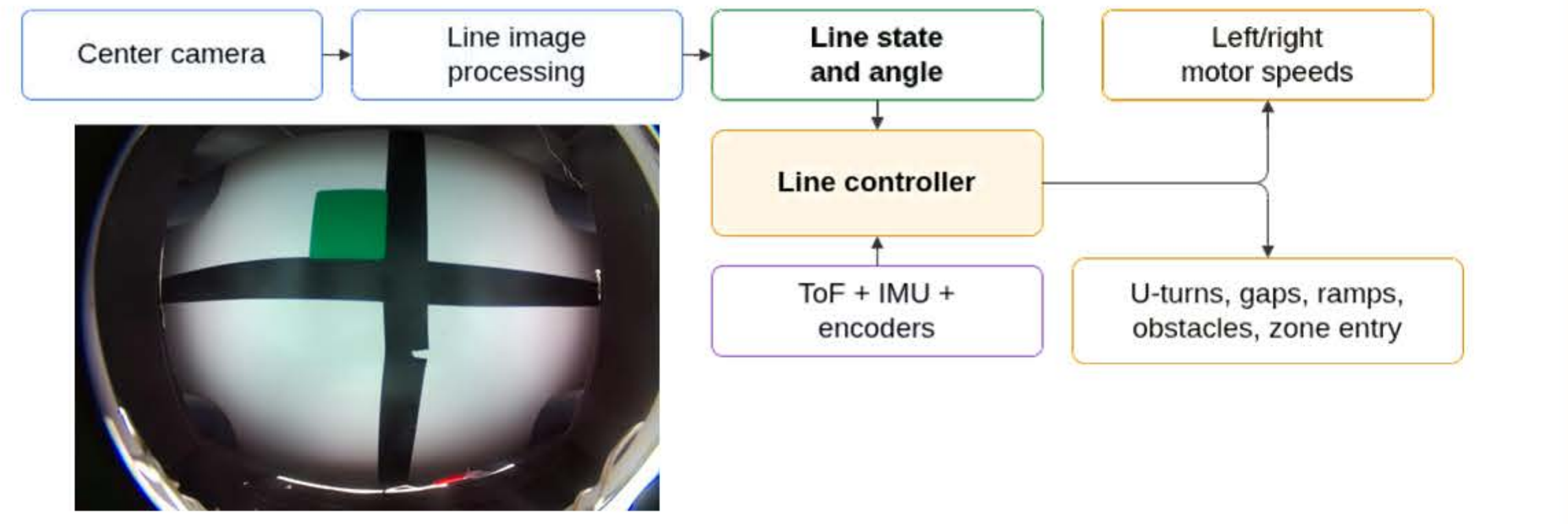
## Main Control Loop

Our non-blocking main control loop updates at an average of 1.24 ms. Every cycle reads a complete sensor snapshot, evaluates the state machine (Line, Zone, or LoP), and applies motor outputs seamlessly without hardware latency.

## Control Flow



## Line Following



## Rescue Zone Behavior

